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Endoscope and endoscope system

Abstract

An endoscope includes, in a distal end frame constituting a distal end portion, an objective optical unit including a voice coil motor that moves a moving lens frame forward and backward in a direction of an optical axis by using a plurality of magnets arranged on a periphery of the moving lens frame, and a plurality of internal components made of magnetic material and arranged around the objective optical unit. At least one of the internal components is arranged at a position where at least a part thereof intersects one of straight lines each connecting an extreme point of one of valleys of a specific isomagnetic line of magnetic fields formed by the magnets and one of two inflection points of the specific isomagnetic line, the two inflection points being located, with the one of the valleys positioned between the two inflection points.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation application of PCT/JP2020/009256 filed on Mar. 4, 2020, the entire contents of which are incorporated herein by this reference.

BACKGROUND OF THE INVENTION
1. Field of the Invention
(1) The present invention relates to an endoscope and an endoscope system that change an optical characteristic of an objective optical system by moving a movable frame by using an electromagnetic actuator.
2. Description of the Related Art
(2) Conventionally, in a field of endoscopes, an endoscope capable of changing an optical characteristic of an objective optical unit provided in a distal end portion of the endoscope by moving a moving lens frame forward and backward in a direction of an optical axis has been proposed and put into practical use.
(3) As an example of an objective optical unit of the endoscope of this type, Japanese Patent Application Laid-Open Publication No. 2015-114651 discloses an optical unit including: a cylindrical fixed portion (fixed frame); a cylindrical movable frame (movable portion) arranged inside the fixed frame; and a voice coil motor capable of relatively moving the movable frame in a direction of an optical axis with respect to the fixed frame by using a coil arranged in the fixed frame and magnets arranged in the movable frame.
(4) In general, in the voice coil motor, the arrangement, the magnetic fields, and the like of the respective magnets are adjusted in order to enable the movable frame to be operated properly in the objective optical unit alone.
SUMMARY OF THE INVENTION
(5) An endoscope according to one aspect of the present invention includes: an objective optical unit including an actuator, the actuator being configured to move a movable frame forward and backward in a direction of an optical axis by using a plurality of magnets; and a plurality of internal components made of magnetic material and arranged around the objective optical unit, the objective optical unit and the plurality of internal components being held in a distal end frame constituting a distal end portion of an insertion portion of the endoscope. At least one of the plurality of internal components is arranged at a position where at least a part of the at least one of the plurality of internal components intersects one of straight lines each connecting an extreme point of one of valleys of a specific isomagnetic line of magnetic fields formed by the plurality of magnets and one of two inflection points of the specific isomagnetic line, the two inflection points being located, with the one of the valleys positioned between the two inflection points.
(6) An endoscope according to another aspect of the present invention includes: an objective optical unit including an actuator, the actuator being configured to move a movable frame forward and backward in a direction of an optical axis by using a plurality of magnets; and a plurality of internal components made of magnetic material and arranged around the objective optical unit, the objective optical unit and the plurality of internal components being held in a distal end frame constituting a distal end portion of an insertion portion of the endoscope. At least one of the plurality of internal components is arranged at a position where at least a part of the at least one of the plurality of internal components intersects one of lines each connecting extreme points of valleys of isomagnetic lines indicating different magnetic flux densities of magnetic fields formed by the plurality of magnets.
(7) In addition, an endoscope system according to one aspect of the present invention includes the

endoscope and an image processing apparatus configured to covert an image pickup signal, which is obtained by image pickup by the endoscope, into an image signal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a schematic configuration view of an endoscope system.

(2) FIG. **2** is a view of an end surface of a distal end portion.

(3) FIG. **3** is a sectional view taken along the line III-III in FIG. **2**.

(4) FIG. **4** is a sectional view taken along the line IV-IV in FIG. **2**.

(5) FIG. **5** is a sectional view taken along the line V-V in FIG. **3**.

(6) FIG. **6** is an explanatory view showing an arrangement relationship between respective internal components and magnetic fields formed by magnets of a voice coil motor.

(7) FIG. **7** is an exploded perspective view of an objective optical unit.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT(S)

(8) Hereinafter, an embodiment of the present invention will be described with reference to drawings. The drawings relate to one embodiment of the present invention and FIG. **1** is a schematic configuration view of an endoscope system.

(9) An endoscope system **1** shown in FIG. **1** includes an endoscope **2**, a control apparatus **3**, and a display apparatus **4**.

(10) The endoscope **2** is configured to be insertable into a subject such as a human body, and to optically observe a predetermined observation site in the subject. Note that the subject into which the endoscope **2** is inserted is not limited to a human body, but may be another living body, or an artificial object such as a machine, a structure, etc.

(11) The endoscope **2** includes an insertion portion **5** configured to be inserted into a subject, an operation portion **6** provided continuously with a proximal end side of the insertion portion **5**, and a universal cord **7** extended from the operation portion **6**.

(12) The insertion portion **5** includes, in the following order from the distal end side toward the proximal end side, a distal end portion **10**, a bending portion **11** configured to be bendable, and a flexible tube portion **12** having flexibility.

(13) Although details will be described later, as shown in FIG. **2**, the distal end portion **10** is provided with: an image pickup unit **20** configured to form an image of a subject on an image pickup device; a plurality of illumination optical units (for example, three illumination optical units: first to third illumination optical units **21**A to **21**C); a treatment instrument channel port **22**; and a gas/liquid feeding nozzle **23**.

(14) A composite cable **25** for transmitting and receiving various signals is connected to the image pickup unit **20** (see FIG. **1**). Further, as shown in FIG. **5**, first to third light guide bundles **26**A to **26**C are connected respectively to the first to third illumination optical units **21**A to **21**C. Furthermore, a treatment instrument channel **27** is connected to the treatment instrument channel port **22**. Furthermore, a gas/liquid feeding tube **28** is connected to the gas/liquid feeding nozzle **23**.

(15) The composite cable **25**, the first to third light guide bundles **26**A to **26**C, the treatment instrument channel **27**, and the gas/liquid feeding tube **28** are passed through the bending portion **11** and the flexible tube portion **12** to be extended to the inside of the operation portion **6**. Note that the respective light guide bundles **26**A to **26**C are merged inside the flexible tube portion **12**, and thereafter extended, as one light guide bundle **26**, to the inside of the operation portion **6**.

(16) The operation portion **6** includes a treatment instrument insertion port **6***a*, an angle operation knob **6***b*, and a zoom lever **6***c*. The treatment instrument insertion port **6***a* configures a proximal end side opening portion of the treatment instrument channel **27**. The angle operation knob **6***b* is configured to perform bending operation of the bending portion **11**. The zoom lever **6***c* is

configured to perform an operation for changing an optical characteristic of the image pickup unit **20**.

(17) The composite cable **25**, the light guide bundle **26**, and the gas/liquid feeding tube **28** that are extended to the inside of the operation portion **6** are inserted through the inside of the universal cord **7**. The universal cord **7** includes, at the proximal end portion thereof, an endoscope connector **8** configured to be detachably connected to the control apparatus **3**.

(18) The endoscope connector **8** is configured to connect the composite cable **25**, the light guide bundle **26**, and the gas/liquid feeding tube **28** to the control apparatus **3**.

(19) The control apparatus **3** includes a processor such as a central processing unit (CPU), and is configured to integrally control the whole endoscope system **1**. The control apparatus **3** includes an image control section **3***a*, a light source control section **3***b*, and a gas/liquid feeding control section **3***c*.

(20) The image control section **3***a* is electrically connected to the image pickup unit **20** and the operation portion **6** through the composite cable **25**. The image control section **3***a* receives an operation signal to the zoom lever **6***c*, to control the optical characteristic of an objective optical unit (to be described later) provided in the image pickup unit **20**. Furthermore, the image control section **3***a* drives and controls the image pickup device (to be described later) of the image pickup unit **20**, and converts an image pickup signal outputted from the image pickup unit **20** into an image signal. An image generated by the conversion in the image control section **3***a* is displayed on the display apparatus **4** such as a monitor. In the present embodiment, the control apparatus **3** includes the image control section **3***a*, to thereby achieve a function as an image processing apparatus.

(21) The light source control section **3***b* is connected to a light source apparatus, not shown, incorporated in the control apparatus **3**. The light source control section **3***b* drives and controls the light source apparatus, to thereby control brightness, etc., of illumination light to be supplied to the first to third illumination optical units **21**A to **21**C through the light guide bundle **26** (the first to third light guide bundles **26**A to **26**C).

(22) The gas/liquid feeding control section **3***c* is connected to a gas/liquid feeding apparatus, not shown, incorporated in the control apparatus **3**. The gas/liquid feeding control section **3***c* drives and controls the gas/liquid feeding apparatus, to thereby feed gas or liquid to the gas/liquid feeding nozzle **23** through the gas/liquid feeding tube **28**.

(23) Next, specific description will be made on the configuration of the distal end portion **10** of the above-described endoscope **2**, with reference to FIGS. **2** to **7**.

(24) The distal end portion **10** includes a distal end frame **30** formed in a substantially columnar shape. The distal end frame **30** is rigid and made of metal such as stainless steel. It is preferable that the magnetization of the distal end frame **30** is reset by performing known thermal processing and the like.

(25) A distal end cover **31**, which forms the distal end surface of the distal end portion **10**, is adhered and fixed to the distal end side of the distal end frame **30**. In addition, the outer circumference of the distal end frame **30** is covered with an outer cover **32**. Furthermore, the outer circumferential portion on the distal end side of the outer cover **32** is fixed to the distal end frame by a thread-wound adhering portion **33**.

(26) The distal end frame **30** is provided with an image pickup unit holding hole **30***a*. The image pickup unit **20** is inserted and fitted in the image pickup unit holding hole **30***a* to be fixed therein by a set screw, not shown, or the like. The distal end of the image pickup unit **20** is exposed outside the distal end cover **31**, to thereby form an observation window **20***a* on the distal end portion **10**.

(27) As shown in FIGS. **4** and **5**, the distal end frame **30** includes a plurality of illumination optical unit holding holes (for example, three illumination optical unit holding holes: first to third illumination optical unit holding holes **30***b*A to **30***b*C) at positions surrounding the periphery of the image pickup unit **20**. The first to third illumination optical units **21**A to **21**C are inserted and fitted

respectively in the first to third illumination optical unit holding holes **30***b*A to **30***b*C.
(28) Each of the first to third illumination optical units **21**A to **21**C includes an illumination lens frame **35** made of metal and formed in a substantially cylindrical shape and a plurality of lenses **36** held in the illumination lens frame **35**.
(29) The distal ends of the first to third illumination optical units **21**A to **21**C are exposed outside the distal end cover **31**, to thereby form illumination windows **21***a*A to **21***a*C, respectively, on the distal end portion **10**.
(30) On the other hand, on the proximal end sides of the first to third illumination optical units **21**A to **21**C, the distal end sides of the first to third light guide bundles **26**A to **26**C to which light guide pipe sleeves **37** made of metal are respectively attached are inserted respectively into the first to third illumination optical unit holding holes **30***b*A to **30***b*C. The first to third light guide bundles **26**A to **26**C are fixed by the light guide pipe sleeves **37** being pressed respectively against the inner walls of the first to third light guide bundles **26**A to **26**C by set screws **38** (see FIG. **5**). The respective illumination optical units **21**A to **21**C are thus optically connected respectively to the light guide bundles **26**A to **26**C, to thereby be capable of applying illumination light supplied from the control apparatus **3** to a subject.
(31) Furthermore, as shown in FIGS. **2** and **5**, the distal end frame **30** is provided with a through hole for channel **30***c*. The distal end of the through hole for channel **30***c* is open as the treatment instrument channel port **22**. A pipe sleeve **40** made of metal is inserted in the through hole for channel **30***c*, and the distal end side of the treatment instrument channel **27** is connected to the pipe sleeve **40**.
(32) As shown in FIGS. **3** and **5**, the distal end frame **30** includes a nozzle insertion hole **30***d* at a position surrounding the periphery of the image pickup unit **20**. The gas/liquid feeding nozzle **23** made of metal is inserted and fitted in the nozzle insertion hole **30***d*. The gas/liquid feeding nozzle **23** is fixed by an adhesive, not shown, or the like, with a spouting port **23***a*, which is provided at the distal end of the nozzle, directed toward the observation window **20***a*. On the other hand, on the proximal end side of the distal end frame **30**, the gas/liquid feeding tube **28** is connected to the gas/liquid feeding nozzle **23**. The gas/liquid feeding nozzle **23** is capable of jetting air and cleaning water supplied from the control apparatus **3** toward the observation window **20***a*.
(33) Next, description will be made on the configuration of the image pickup unit **20** provided in the distal end portion **10**.
(34) As shown in FIGS. **3** to **5**, and **7**, the image pickup unit **20** includes an image pickup device unit **45** and an objective optical unit **46** provided continuously with the distal end side of the image pickup device unit **45**.
(35) The image pickup device unit **45** includes an image pickup device holding frame **50**. In the image pickup device holding frame **50**, the front face side of a solid-state image pickup device **51** constituted of CCD, CMOS, and the like, is held through an optical member **52** such as a cover glass. An image pickup device substrate **53** is electrically connected to the rear face side of the solid-state image pickup device **51**. On the image pickup device substrate **53**, various control circuits and the like are mounted. Although not shown, various cables, which are branched from the composite cable **25**, are electrically connected to the image pickup device substrate **53**.
(36) The objective optical unit **46** includes a front-group-lens frame **55**, a rear-group-lens frame **56**, a coil holding frame **57**, a moving lens frame **58**, and a sensor holding frame **59**. The coil holding frame **57** is disposed between the front-group-lens frame **55** and the rear-group-lens frame **56**. The moving lens frame **58** is a movable frame arranged slidably in the coil holding frame **57**. The sensor holding frame **59** integrally holds the front-group-lens frame **55**, the rear-group-lens frame **56**, and the coil holding frame **57**. The objective optical unit **46** is an optical unit capable of changing the optical characteristic of the objective optical system by moving the moving lens frame **58** forward and backward in a direction of an optical axis O by using a voice coil motor **60** (see FIG. **7**) to be described later.

(37) The front-group-lens frame **55** is configured of a frame body formed in a substantially cylindrical shape. The front-group-lens frame **55** is formed such that the outer diameter on the distal end side is larger than the outer diameter on the proximal end side. The outer diameter on the distal end side thus differs in size from the outer diameter on the proximal end side, to thereby form a step portion **55***a* at a midway part of the outer circumferential surface of the front-group-lens frame **55**. The step portion **55***a* is set as a contact surface for positioning the front-group-lens frame **55** with respect to the sensor holding frame **59**.
(38) In addition, the front-group-lens frame **55** holds inside thereof a front-group lens **65** constituted of a plurality of fixed lenses. The front-group lens **65** constitutes the objective optical system of the objective optical unit **46**.
(39) The rear-group-lens frame **56** is configured of a frame body formed in a substantially cylindrical shape. The rear-group-lens frame **56** is formed such that the outer diameter on the distal end side is smaller than the outer diameter on the proximal end side. The outer diameter on the distal end side thus differs in size from the outer diameter on the proximal end side, to thereby form a step portion **56***a* at a midway part of the outer circumferential surface of the rear-group-lens frame **56**. The step portion **56***a* is set as a contact surface for positioning the rear-group-lens frame **56** with respect to the coil holding frame **57**.
(40) In addition, the rear-group-lens frame **56** holds inside thereof a rear-group lens **66** constituted of a plurality of fixed lenses. The rear-group lens **66** constitutes the objective optical system of the objective optical unit **46**.
(41) The coil holding frame **57** is configured of a frame body formed in a substantially cylindrical shape. The coil holding frame **57** includes reduced thickness portions **57***a* extending in the direction of the optical axis O. The reduced thickness portions **57***a* are formed at four positions at 90 degree intervals around the optical axis O (see FIG. **7**).
(42) Two coils **67***f*, **67***r* constituting the voice coil motor **60** are arranged on the outer circumferential surface of the coil holding frame **57**. The two coils **67***f*, **67***r* are arranged side by side in the direction of the optical axis O.
(43) The moving lens frame **58** is configured of a frame body formed in a substantially cylindrical shape. The moving lens frame **58** holds inside thereof a moving group lens **58***a* constituted of one lens or two or more lenses. The moving group lens **58***a* constitutes the objective optical system of the objective optical unit **46**.
(44) A pair of recessed portions **68***f*, **68***r* is provided at each of four positions on the outer circumferential surface of the moving lens frame **58** so as to be located at each 90 degree point around the optical axis O. The recessed portion **68***f* and the recessed portion **68***r* in each pair are arranged side by side in the direction of the optical axis O. The recessed portion **68***f* and the recessed portion **68***r* in each pair respectively hold two magnets **69***f* and **69***r*, which constitute the voice coil motor **60**, such that the two magnets are arranged side by side in the direction of the optical axis O.
(45) Each of the magnets **69***f*, **69***r* is arranged such that a part of each of the magnets protrudes from each of the recessed portions **68***f*, **68***r* in a radially outward direction of the moving lens frame **58**. In addition, each of the magnets **69***f*, **69***r* is magnetized so as to have a polarity in the thickness direction (that is, in the radial direction of the moving lens frame **58**). The magnets **69***f*, **69***r* in the present embodiment are arranged respectively in the recessed portions **68***f*, **68***r* such that each magnet has an S-pole in the radially outward direction of the moving lens frame **58**, and an N-pole in the radially inward direction of the moving lens frame **58**, and the magnets **69***f*, **69***r* are fixed respectively to the recessed portions **68**, **68***r* by an adhesive.
(46) Among the four pairs of recessed portions **68***f*, **68***r* disposed on the moving lens frame **58**, one pair of recessed portions **68***f*, **68***r* have a key **70** therebetween. The key **70** protrudes in the radially outward direction from between the one pair of magnets **69***f*, **69***r* arranged in the one pair of recessed portions **68***f*, **68***r*. The key **70** is inserted into any one of the four reduced thickness

portions **57***a* when the moving lens frame **58** is inserted in the coil holding frame **57**. Such insertion of the key **70** into one of the reduced thickness portions **57***a* restricts the rotation of the moving lens frame **58** around the optical axis O. In other words, the moving lens frame **58** is housed inside the coil holding frame **57**, with the rotation around the optical axis O being restricted by the key **70** and the movement in the direction of the optical axis O being allowed.

(47) The sensor holding frame **59** is configured of a frame body formed in a substantially cylindrical shape. The sensor holding frame **59** includes, on the inner circumference on the distal end side thereof, an inward flange **59***a*.

(48) The sensor holding frame **59** includes, on one side of the outer circumferential portion thereof, a sensor holding groove **75**. The sensor holding groove **75** includes, in a part on the distal end side thereof, a through hole **75***a* penetrating the inside and outside of the sensor holding frame **59**.

(49) A sensor substrate **76** is arranged inside the sensor holding groove **75**. The sensor substrate **76** is provided with a Hall element **77** for detecting magnetic fields. The Hall element **77** is disposed on a surface facing the through hole **75***a*. Various cables **25***a* branched from the composite cable **25** are connected to the sensor substrate **76**. The sensor holding groove **75** is provided with a biasing plate **78**, which is configured of a metal plate, on a radially outward position with respect to the sensor substrate **76**.

(50) The sensor holding groove **75** in which the sensor substrate **76** and the biasing plate **78** are thus housed is closed by a lid body **79**.

(51) The sensor holding frame **59** holds, on the distal end side thereof, the front-group-lens frame **55**. Specifically, the front-group-lens frame **55** is fixed to the sensor holding frame **59** by adhering or the like, with the proximal end side of the front-group-lens frame **55** being inserted into the sensor holding frame **59**. The step portion **55***a* of the front-group-lens frame **55** is in contact with the distal end surface of the inward flange **59***a* of the sensor holding frame **59**, with an adhesive, not shown, interposed therebetween. The front-group-lens frame **55** is thus positioned in the direction of the optical axis O with respect to the sensor holding frame **59**.

(52) In addition, the coil holding frame **57** is held inside the sensor holding frame **59**. Specifically, the coil holding frame **57** is fixed by adhering or the like, with the distal end surface thereof being in contact with the proximal end surface of the inward flange **59***a* of the sensor holding frame **59**.

(53) The positioning of the coil holding frame **57** around the optical axis O is determined such that one pair of magnets **69***f*, **69***r* faces the Hall element **77**. The one pair of magnets **69***f*, **69***r* is among the plurality of pairs (four pairs in the present embodiment) of magnets **69***f*, **69***r* on the moving lens frame **58** housed in the coil holding frame **57**. With such a configuration, the Hall element **77** is capable of detecting a position of the moving lens frame **58** in the direction of the optical axis O based on a change in the magnetic fields received from the one pair of magnets **69***f*, **69***r*. In addition, the one pair of magnets **69***f*, **69***r* is attracted toward the biasing plate **78**, to thereby suppress the backlash of the moving lens frame **58** in the coil holding frame **57**. Such a configuration achieves a stable sliding performance of the moving lens frame **58** and a stable optical characteristic of the objective optical system.

(54) Furthermore, the sensor holding frame **59** holds, inside thereof, the rear-group-lens frame **56** at a position on the proximal end side with respect to the coil holding frame **57**. Specifically, the rear-group-lens frame **56** is fixed by an adhesive or the like, with the distal end side thereof being inserted in the coil holding frame **57** and the proximal end side thereof being inserted in the sensor holding frame **59**. The rear-group-lens frame **56** is positioned in the direction of the optical axis O with respect to the coil holding frame **57** by the step portion being brought into contact with the proximal end surface of the coil holding frame **57**.

(55) The proximal end side of the rear-group-lens frame **56** thus held in the sensor holding frame **59** is coupled with the distal end side of the image pickup device holding frame **50**. With such a configuration, an image of the subject is formed on the solid-state image pickup device **51** through the objective optical system of the objective optical unit **46**.

(56) In the endoscope system **1** thus configured, w % ben the zoom lever **6***c* of the endoscope **2** is operated by a user or another person, the image control section **3***a* of the control apparatus **3** performs energization control on the coils **67***f*, **67***r* in accordance with the operation state of the zoom lever **6***c*. The magnets **69***f*, **69***r* receive the magnetic fields generated in the coils **67***f*, **67***r* by the energization control, to move the moving lens frame **58** in the direction of the optical axis O. The Hall element **77** detects a change in the magnetic fields according to the movement of the moving lens frame **58** in the direction of the optical axis O by the magnets **69***f*, **69***r*, and outputs the detected change in the magnetic fields, as a feedback signal, to the image control section **3***a*. In response to the feedback signal, the energization of the coils **67***f*, **67***r* is feedback controlled, and the moving lens frame **58** is controlled at a position on the optical axis O in accordance with the operation state of the zoom lever **6***c*.
(57) As shown in FIG. **6**, the objective optical unit **46** of the image pickup unit **20** forms the magnetic fields around the objective optical unit **46** by the four pairs of magnets **69***f*, **69***r*. Isomagnetic lines of the magnetic fields shown in FIG. **6** indicate a distribution of the magnetic fields of the objective optical unit **46** alone before the objective optical unit **46** is incorporated in the distal end frame **30**.
(58) In order to prevent the magnetic fields from being changed due to an influence of another metal (magnetic material), various internal components are arranged in the distal end portion **10** at positions where the internal components hardly have an influence on the magnetic fields formed by the respective magnets **69***f*, **69***r* and the internal components are in the vicinity of the image pickup unit **20**.
(59) Specifically, as shown in FIG. **6**, for example, the various internal components made of magnetic material are arranged so as not to substantially intrude inside a specific isomagnetic line **80** (in the present embodiment, 100 (mT), for example) set as a rough guide line. The specific isomagnetic line **80** indicates a lower limit value of the magnetic intensity which is considered to have an influence on the operating performance of the voice coil motor **60**. The lower limit value is determined appropriately based on experiments, simulations, and the like, depending on the type of the objective optical unit **46**. Note that even in the case where the internal components intrude inside the specific isomagnetic line **80**, the intrusion can be allowed if the intruding amount is so slight as not to have an influence on the magnetic fields.
(60) In order to bring the various internal components close to the image pickup unit **20** while satisfying the above-described condition, it is preferable that at least one of the various internal components is arranged such that at least a part thereof is located on any one of lines (first to fourth lines **83***a* to **83***d*) each connecting extreme points of valleys of the isomagnetic lines indicating different magnetic flux densities of the magnetic fields. Alternatively, it is preferable that at least one of the various internal components is arranged at a position intersecting at least one of straight lines (first to eighth straight lines **84***a* to **84***g*) each connecting one of extreme points **81***a* to **81***d* of the valleys of the specific isomagnetic line **80** and one of two inflection points located with one of the valleys positioned therebetween, among the inflection points **82***a* to **82***h* of the specific isomagnetic line **80**. With such an arrangement satisfying at least one of the above-described conditions, at least one of the various internal components can be arranged along one of the valleys of the specific isomagnetic line **80**, to thereby enable the internal component to be arranged as close to the objective optical unit **46** as possible, while eliminating the influence on the magnetic fields.
(61) Here, the extreme point of each of the valleys of the isomagnetic lines refers to the extreme point of a part of each of the isomagnetic lines, the part having a shape protruding toward between one pair of magnets **69***f*, **69***r* and adjoining pair of magnets **69***f*, **69***r*. In addition, the first to fourth lines **83***a* to **83***d* each connecting the extreme points of the valleys of the respective isomagnetic lines may be approximation straight lines or may be spline curves. The four pairs of magnets **69***f*, **69***r* are arranged on the outer circumference of the moving lens frame **58** of the present

embodiment. As a result, the valleys of each of the isomagnetic lines are formed respectively at four positions. Therefore, the first to fourth lines **83***a* to **83***d* each connecting the valleys of the respective isomagnetic lines are set around the objective optical unit **46**. In addition, the first to eighth straight lines **84***a* to **84***h* each connecting the extreme point of one of the valleys and one of the inflection points of the specific isomagnetic line **80** are set.

(62) In the present embodiment, the first illumination optical unit **21**A is arranged at a position where at least a part thereof intersects the first line **83***a*. Furthermore, the first illumination optical unit **21**A is arranged at a position where at least a part thereof intersects the first straight line **84***a* and a large part (for example, half or more of the cross-sectional area) thereof is located within a region between the two straight lines specified by the same extreme point (extreme point **81***a*). By satisfying these conditions, the first illumination optical unit **21**A is arranged along the one of the valleys of the specific isomagnetic line **80**.

(63) Such an arrangement achieves a configuration in which the illumination lens frame **35**, the light guide pipe sleeve **37** and the set screw **38**, which are the various components (internal components) related to the first illumination optical unit **21**A and made of magnetic material, are not arranged substantially inside the specific isomagnetic line **80**, but are arranged closely to the objective optical unit **46** (image pickup unit **20**).

(64) Note that, in the present embodiment, the first illumination optical unit **21**A is arranged such that a part of the illumination lens frame **35** and a part of the light guide pipe sleeve **37** intersect the specific isomagnetic line **80**. However, the intrusion amount (cross-sectional area) of the illumination lens frame **35** and the light guide pipe sleeve **37** into the inside of the specific isomagnetic line **80** is so slight as not to have a great influence on the magnetic fields.

(65) In the present embodiment, it is preferable that an intruding area Sa of the internal components, which are made of magnetic material, into the inside of the specific isomagnetic line **80** is set such that an estimated value E shown below by the equation (1) is less than a threshold Eth throughout the overall area in the direction of the optical axis O.

$E=(Sa/S)\times P \qquad (1)$

(66) Note that, in the equation (1), S indicates the cross-sectional area of the internal components made of magnetic material, and P indicates a magnetic permeability of the internal components made of magnetic material.

(67) Such an arrangement enables a part of the internal components to be arranged inside the specific isomagnetic line **80** without having a great influence on the magnetic fields of the voice coil motor **60**. Note that the threshold value Eth has been obtained in advance based on experiments, simulations, and the like.

(68) The second illumination optical unit **21**B is arranged at a position where at least a part thereof intersects the second line **83***b*. By satisfying such a condition, the second illumination optical unit **21**B is arranged along one of the valleys of the specific isomagnetic line **80**.

(69) Such an arrangement achieves a configuration in which the illumination lens frame **35**, the light guide pipe sleeve **37**, and the set screw **38**, which are the various components (internal components) related to the second illumination optical unit **21**B and made of magnetic material, are not arranged substantially inside the specific isomagnetic line **80** but are arranged closely to the objective optical unit **46** (image pickup unit **20**).

(70) The third illumination optical unit **21**C is arranged at a position where at least a part thereof intersects the third line **83***c*. By satisfying such a condition, the third illumination optical unit **21**C is arranged along one of the valleys of the specific isomagnetic line **80**.

(71) Such an arrangement achieves a configuration in which the illumination lens frame **35**, the light guide pipe sleeve **37**, and the set screw **38**, which are the various components (internal components) related to the third illumination optical unit **21**C and made of magnetic material, are not arranged substantially inside the specific isomagnetic line **80** but arranged closely to the objective optical unit **46** (image pickup unit **20**).

(72) Furthermore, the gas/liquid feeding nozzle **23** is arranged at a position where at least a part thereof intersects the second straight line **84***b* and a large part thereof is located within a region between the two straight lines specified by the same extreme point (extreme point **81***a*). By satisfying these conditions, the gas/liquid feeding nozzle **23** is arranged along one of the valleys of the specific isomagnetic line **80**.
(73) Note that, in the example shown in FIG. **6**, the treatment instrument channel port **22** is away from the objective optical unit **46** on the layout. Therefore, it is not especially necessary to satisfy the above-described conditions, however, if there is a need for arranging the treatment instrument channel port **22** in the vicinity of the objective optical unit **46**, the treatment instrument channel port **22** is preferably arranged at a position where at least a part thereof intersecting any one of the first to fourth straight lines **84***a* to **84***d*.
(74) According to such an embodiment, the endoscope **2** holds, in the distal end frame **30** constituting the distal end portion **10**, the objective optical unit **46** and the plurality of internal components made of magnetic material and arranged around the objective optical unit **46**, and the objective optical unit **46** includes the voice coil motor **60** configured to move the moving lens frame **58** forward and backward in the direction of the optical axis O by using the plurality of magnets **69***f*, **69***r* arranged on the periphery of the moving lens frame **58**. In such an endoscope **2**, at least one of the plurality of internal components is arranged at a position where at least a part of the at least one of the plurality of internal components intersects any one of the first to fourth lines **83***a* to **83***d* each connecting the extreme points of the valleys of the isomagnetic lines indicating the different magnetic flux densities of the magnetic fields formed by the plurality of magnets **69***f*, **69***r*. Such an arrangement ensures the operation performance of the moving lens frame **58** with an inexpensive configuration, without increasing the diameter size of the distal end portion **10**.
(75) In other words, at least one of the plurality of internal components is arranged at a position where at least a part of the at least one of the plurality of internal components intersects any one of the first to fourth lines **83***a* to **83***d* each connecting the extreme points of the valleys of the isomagnetic lines, to thereby enable the at least one internal component to be efficiently arranged along one of the valleys of the specific isomagnetic line **80**. Therefore, the internal components can be arranged in the vicinity of the objective optical unit **46** while suppressing the influence of the internal components on the magnetic fields. In addition, it is possible to suppress the increase in the diameter size of the distal end portion **10**. The internal components are thus arranged without having an influence on the magnetic fields. Therefore, it is not necessary to change the arrangement of the magnets **69***f*, **69***r* of the voice coil motor **60** provided in the objective optical unit **46** and the magnetic fields formed by the magnets depending on the type or the like of the endoscope **2**. As a result, the operation performance of the moving lens frame **58** can be ensured with the inexpensive configuration.
(76) In addition, at least one of the plurality of internal components is arranged at a position where at least a part of the at least one of the plurality of internal components intersects any one of the first to eighth straight lines **84***a* to **84***h* each connecting one of the extreme points **81***a* to **81***d* of the valleys of the specific isomagnetic line **80** and one of the two inflection points located with the one of the valleys positioned therebetween, among the plurality of inflection points **82***a* to **82***h* of the specific isomagnetic line **80**. Also in such a case, it is possible to efficiently arrange the at least one internal component along one of the valleys of the specific isomagnetic line **80**.
(77) Note that the present invention is not limited to the above-described embodiment, but various modifications and changes are possible, and such modifications and changes are also within the technical range of the present invention.
(78) In the above-described embodiment, description has been made on the configuration in which all the three illumination optical units **21**A to **21**C are arranged at the positions each intersecting any one of the first to fourth lines **83***a* to **83***d*, for example. However, the present invention is not limited to the configuration. Depending on the layout on the distal end portion **10**, one or two

illumination optical units can be arranged at a position or positions each intersecting any one of the first to fourth lines **83***a* to **83***d*.

(79) In addition the number of the illumination optical units to be arranged in the distal end portion **10** is not limited to three. One or two illumination optical units, or four or more illumination optical units may be arranged.

(80) In the above-described embodiment, description has been made on one example in which each of the illumination optical units is arranged at the position intersecting any one of the first to fourth lines **83***a* to **83***d*. However, for example, another internal component such as the treatment instrument channel port **22**, the gas/liquid feeding nozzle **23**, or the like, can be arranged at a position intersecting any one of the first to fourth lines **83***a* to **83***d*.

(81) Furthermore, it is needless to say that the number of magnets constituting the voice coil motor **60** is not limited to that in the above-described embodiment.

Claims

1. An endoscope comprising: an objective optical unit comprising: a lens; a lens frame holding the lens; and an actuator configured to move the lens frame in a direction of an optical axis, the actuator including a plurality of magnets and a coil; and a plurality of internal components made of magnetic material and arranged around the objective optical unit, wherein the plurality of internal components are other than portions comprising the actuator; at least one of the plurality of internal components is arranged at a position where at least a part of the at least one of the plurality of internal components intersects a straight line connecting an extreme point of one of valleys of a specific isomagnetic line of magnetic fields formed by the plurality of magnets and one of two inflection points of the specific isomagnetic line, the one of the valleys positioned between the two inflection points; and each of the plurality of internal components are fixed relative to the lens frame.
2. The endoscope according to claim 1, wherein at least one of the plurality of internal components is arranged at a position where at least a part of the at least one of the plurality of internal components intersects one of lines each connecting extreme points of valleys of isomagnetic lines indicating different magnetic flux densities of the magnetic fields formed by the plurality of magnets.
3. The endoscope according to claim 1, wherein the plurality of internal components include an illumination optical unit and a component related to the illumination optical unit.
4. The endoscope according to claim 1, wherein the plurality of internal components include a gas/liquid feeding nozzle and a component related to the gas/liquid feeding nozzle.
5. The endoscope according to claim 1, wherein the plurality of internal components include a treatment instrument channel port and a component related to the treatment instrument channel port.
6. An endoscope system comprising: the endoscope according to claim 1; and an image processing apparatus configured to convert an image pickup signal obtained by image pickup by the endoscope into an image signal.
7. The endoscope according to claim 1, wherein the plurality of internal components comprise one or more of an illumination lens frame, a light guide pipe sleeve and a set screw.
8. The endoscope according to claim 1, wherein the actuator further comprises: the coil arranged around the plurality of magnets; and a coil frame for holding the coil, the coil frame located between the plurality of magnets and the coil.
9. An endoscope comprising: an objective optical unit comprising: a lens; a lens frame holding the lens; and an actuator connected to the lens frame, the actuator configured to move the lens frame in a direction of an optical axis, the actuator including a plurality of magnets and a coil; and a plurality of internal components made of magnetic material and arranged around the objective

optical unit, wherein the plurality of internal components are other than portions comprising the actuator; at least one of the plurality of internal components is arranged at a position where at least a part of the at least one of the plurality of internal components intersects a straight line connecting extreme points of valleys of isomagnetic lines indicating different magnetic flux densities of magnetic fields formed by the plurality of magnets; and each of the plurality of internal components are fixed relative to the lens frame.

10. The endoscope according to claim 9, wherein the plurality of internal components include an illumination optical unit and a component related to the illumination optical unit.

11. The endoscope according to claim 9, wherein the plurality of internal components include a gas/liquid feeding nozzle and a component related to the gas/liquid feeding nozzle.

12. The endoscope according to claim 9, wherein the plurality of internal components include a treatment instrument channel port and a component related to the treatment instrument channel port.

13. An endoscope system comprising: the endoscope according to claim 6; and an image processing apparatus configured to convert an image pickup signal obtained by image pickup by the endoscope into an image signal.

14. The endoscope according to claim 9, wherein the plurality of internal components comprise one or more of an illumination lens frame, a light guide pipe sleeve and a set screw.

15. The endoscope according to claim 9, wherein the actuator further comprises: the coil arranged around the plurality of magnets; and a coil frame for holding the coil, the coil frame located between the plurality of magnets and the coil.

16. An endoscope comprising: an objective optical unit held by the frame, the objective optical unit comprising: a lens; a lens frame holding the lens; and an actuator connected to the lens frame, the actuator configured to move the lens frame in a direction of an optical axis, the actuator including a plurality of magnets; and a plurality of internal components made of magnetic material and arranged around the objective optical unit, wherein the plurality of internal components comprises one or more of an illumination lens frame, a light guide pipe sleeve and a set screw; at least one of the plurality of internal components is arranged at a position where at least a part of the at least one of the plurality of internal components intersects a straight line connecting an extreme point of one of valleys of a specific isomagnetic line of magnetic fields formed by the plurality of magnets and one of two inflection points of the specific isomagnetic line, the one of the valleys positioned between the two inflection points; and each of the plurality of internal components are fixed relative to the lens frame.

17. The endoscope according to claim 16, wherein the actuator further comprises: a coil arranged around the plurality of magnets; and a coil frame for holding the coil, the coil frame located between the plurality of magnets and the coil.